

Toward Task-Aware Reuse of Cellular Molecular Receivers: History-Aware Prediction on Experimental FGF2–ERK Data

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Abstract—Chemical commands delivered to a living receiver can change how it responds to later commands. Using public experimental FGF2–ERK single-cell trajectories rather than artificial bit labels, we test whether measured stimulation history improves prediction of a cellular receiver’s reporter and whether such information could shorten repeated use. A provenance-checked pilot reduced the 10-min-ahead root mean squared error (RMSE) by 8.25% relative to a current-input/response/clock baseline on an already inspected test. A pre-registered expansion to 14 unused conditions (22 conditions, 1,565 cells) confirmed an 8.8% gain over that baseline on an unseen 10-min-pulse protocol, but only 0.7% over naive persistence. We then executed the original authors’ B3 biochemical model from their files, reproducing their posterior predictions to within 10^{-5} . In a post hoc reference comparison, a matched-feedback B3 forecaster alone only equalled persistence, whereas adding causal-history features to it reduced the 10-min RMSE on the unseen protocol by a further 8.2%, at four of four concentrations. Finally, we define task-dependent receiver readiness and, in simulation on B3, find that after 25 ng/ml commands readiness is nearly a function of the current reporter, the candidate waits of 6–48 min never suffice, and a history-aware waiting policy reaches 90% success about 8 min faster than the matching fixed wait, mainly by filtering noisy observations. The results support history- and mechanism-informed prediction, not a new biological mechanism or measured task reliability.

Index Terms—Molecular communication, cellular receivers, chemical signaling, history-aware prediction, pre-registration, FGF2, ERK.

I. INTRODUCTION

An application of molecular communication (MC) can require a chemical signal to produce a specified response in a living cell. Successful reception then means more than recovering a symbol: the receiver must execute the requested response at the right time. A previous exposure may influence the next response through transport persistence, receptor dynamics, intracellular feedback, or changes in cellular state. For a sequence of commands, an operational question is therefore when the receiver can be used again, and at what chemical and temporal cost.

We ground this question in recorded biology rather than artificial labels: FGF2 stimulation of PC-12 cells, with ERK activity measured by a FRET reporter [1]; ERK stimulation patterns are known to shape cellular outcomes [2]. The reporter is treated as a continuous receiver output, and no alphabet or cell-fate label is assigned retroactively.

Prior work bounds what a generic learning-based controller could claim. The FGF2 data were originally used to infer signaling architecture, including receptor interactions and feedback [1]; their nontrivial washout response is established biology, not a newly discovered memory mechanism, and frequency-dependent ERK responses are also known [2]. Postiglione *et al.* controlled mammalian gene expression and signaling with automated chemical delivery and feedback [3], and Lugagne *et al.* used a learned dynamical model for model predictive control (MPC) of thousands of single cells [4]. Nonlinear prediction, response feedback and MPC are therefore substantial precedents; the extension considered here is the feasibility and cost of a *next task conditioned on previous tasks*, not chemical feedback control itself.

Within MC, semantic systems already learn task-relevant representations [5], [6], interpretable reaction–diffusion learning has been proposed [7], experimental yeast pheromone links characterize biological receiver noise and detection [8], genetic-circuit receivers are designed computationally [9], Barros and Dey control calcium signaling and its refractory behavior [10], CELLEC adapts error control to modeled cellular noise [11], and long-term experimental MC shows that reuse and resource tradeoffs require explicit engineering [12]. The candidate distinction to test is a calibrated, sequence-level exposure/time–reliability tradeoff attributable to receiver history beyond a matched-information mechanistic estimator.

The narrower question is therefore whether a measured chemical-response history improves prediction beyond accessible present-state information, and whether that information can eventually reduce exposure or completion time for *repeated tasks at matched reliability*. This draft makes four bounded contributions: (i) an audited experimental input–response pilot with a pre-registered expansion that supplies unseen protocols; (ii) a quantification of causal-history features against current-state models and naive persistence; (iii) an execution of the original B3 model, accepted against the authors’ exports, that provides a matched-feedback mechanistic reference and a nested history-residual comparison; and (iv) a task-dependent readiness criterion with a controller specification, examined in simulation on B3. No result concerns measured task reliability.

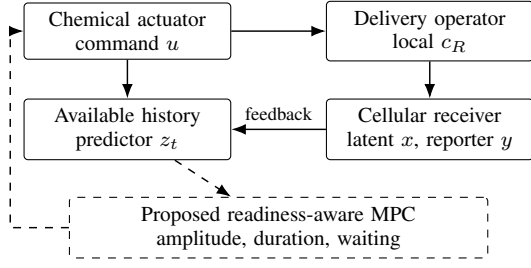


Fig. 1. MC interpretation. The completed analysis predicts the reporter from commands and available observations; local delivery and latent states are not calibrated, and the dashed controller is examined only in simulation (Sec. VI-E).

II. CELLULAR MC MODEL AND TASK READINESS

A. Actuation, Delivery, and Observation

Let $u(t) \geq 0$ denote the commanded FGF2 concentration in ng/ml. Delivery to a receiver is represented abstractly as

$$c_R(t) = \mathcal{T}_\vartheta[u](t), \quad \dot{x}_j = f_{\theta_j}(x_j, c_R), \quad y_j = g(x_j) + \nu_j, \quad (1)$$

where c_R is the local concentration, x_j the unobserved state of cell j , and y_j its normalized reporter. The operator $\mathcal{T}_\vartheta$ can include delivery lag and washout; neither it nor c_R is calibrated here. The FRET ratio is not a molecular count, receptor occupancy, or cell-fate measurement. For deployment, $\mathcal{H}_t$ contains executed commands and observations whose feedback has arrived by time t ; the offline analysis assumes each processed observation is available at its recorded time, an assumption a live experiment must replace with logged latency.

B. Task-Dependent Readiness: Proposed Definition

A prospective command m specifies an output band $\mathcal{Y}_m(t)$, a hold interval with acquisition times $\mathcal{I}_m$, a deadline, and any exclusion interval for false activation, all fixed by a separately calibrated application. At the measurement resolution, task success is

$$S_m = \mathbf{1}\left\{\bigcap_{t_i \in \mathcal{I}_m} \{y_j(t_i) \in \mathcal{Y}_m(t_i)\} \cap \mathcal{C}_m\right\}, \quad (2)$$

where $\mathcal{C}_m$ is the event that deadline and exclusion requirements hold. Individual-cell and population criteria differ and must be chosen before a policy is evaluated. For a candidate future command sequence $\mathbf{a}$, readiness is

$$R_t(m, \mathbf{a}) = \mathbb{P}(S_m = 1 \mid \mathcal{H}_t, \mathbf{a}). \quad (3)$$

Readiness is specific to a requested response and deadline: returning to a reporter baseline need not imply readiness for every next task, and successful delivery may not require complete biochemical reset. These are hypotheses to evaluate. The present data have no prespecified $\mathcal{Y}_m$ or task labels, so Eq. (2) is not applied retrospectively, and a point predictor's RMSE is not a calibrated estimate of Eq. (3).

III. HISTORY-AWARE PREDICTION

We fit separate predictors for horizons $h = 2$ and 10 min. All learned predictors receive the same known future command: its exposure in successive 2-min bins, the concentration at $t + h$, and the horizon. Future reporter values are targets only, so this is conditional forecasting under recorded protocols, not evaluation of counterfactual commands. The input-response (IR) set contains $u_t, u_t^2, y_t, y_t^2, u_t y_t$ and the future-command features; comparators add the clock t, t^2 or the slope $(y_t - y_{t-2})/2$. The causal-history set adds this slope, the past exposure $\int_0^t u(s) ds$, and input and response filters with scales $\tau \in \{2, 10, 60\}$ min,

$$q_\tau(t) = \frac{1}{\tau} \int_0^t e^{-(t-s)/\tau} u(s) ds, \quad (4)$$

$$r_{\tau,i} = e^{-\Delta_i/\tau} r_{\tau,i-1} + (1 - e^{-\Delta_i/\tau}) y_i,$$

with $r_{\tau,0} = y_0$ and $\Delta_i = t_i - t_{i-1}$; q_τ is evaluated exactly for the piecewise-constant command. The scales are computational choices, not identified rate constants. A history-plus-clock set adds t, t^2 to the history features, giving a nested ablation of the IR-plus-clock comparator. Prediction starts after a fixed 10-min observed warmup; pre-export biological history is unknown.

Let d index training condition blocks with N_d rows each. Row weights $w_i = 1/(D_{\text{tr}} N_{d(i)})$ give every block equal weight. With features standardized by weighted training statistics, standardized row a_i and weighted target mean $\bar{y}$,

$$\hat{\beta}_\lambda = \arg \min_{\beta} \sum_{i \in \text{train}} w_i (y_{i+h} - \bar{y} - a_i^\top \beta)^2 + \lambda \|\beta\|_2^2, \quad (5)$$

with the unpenalized intercept $\bar{y}$. The penalty is selected by condition-equal validation MSE, separately per predictor and horizon; validation blocks are never added to training. The small feature set tests whether history carries usable information before a more complex architecture is justified.

As a mechanistic reference we execute the original authors' B3 model with their 1,000 posterior parameter samples (Sec. V). With $f_s(t)$ the noise-free reporter of sample s under the recorded command, the matched-feedback forecaster M2 weights the samples by each cell's past reporter,

$$w_s(t) \propto \exp \left[- \sum_{t_i \leq t} (y_i - f_s(t_i))^2 / (2\sigma^2) \right], \quad (6)$$

$$\hat{y}_{t+h} = y_t + \sum_s w_s(t) [f_s(t+h) - f_s(t)],$$

a fixed-parameter particle approximation of the B3 belief that uses the same information as the learned predictors. M3 adds the M2 increment as one feature to the causal-history ridge, so it nests both causal history (M1) and M2; IR plus clock is M0. The scale σ is selected on validation from seven values. The samples express population-mean uncertainty, not cell heterogeneity.

IV. PROPOSED RECEIVER-REUSE CONTROLLER

The proposed extension uses actions $a_k = (A_k, d_k, w_k)$: command amplitude, pulse duration, and waiting/washout time, within an experimentally calibrated set $\mathcal{A}_{\text{cal}}$ that also

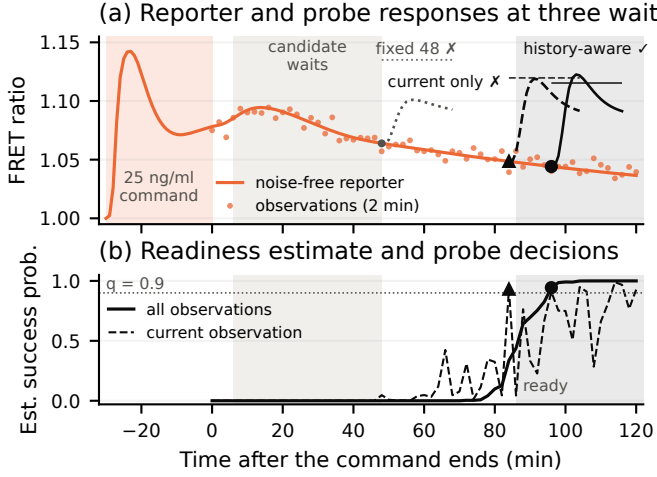


Fig. 2. Readiness-aware waiting simulated on one B3 posterior sample after a 30-min, 25 ng/ml command. (a) Reporter, noisy observations, and 20-min responses to a 5-min probe at three waits; segments mark the required rise. (b) Success probability estimated from all or only the current observation; a policy probes when it reaches q . Gray: this receiver is ready. Chosen by a fixed rule from the 194 of 1,000 samples where the current-only policy probed early and failed and the history-aware one succeeded (321 others: current-only later).

encodes actuator bounds, task order and deadlines. Chemical washout and cellular recovery can differ and should be measured separately. When cells share a chamber and one command, the policy must optimize a prespecified population criterion. For ordered tasks with $S_{\text{seq}} = \prod_{k=1}^K S_{m_k}$, a candidate planning problem is

$$\begin{aligned} \min_{\mathbf{a} \in \mathcal{A}_{\text{cal}}} \quad & \eta_E \mathbb{E}[E(\mathbf{a}) | \mathcal{H}_t] + \eta_T \mathbb{E}[T(\mathbf{a}) | \mathcal{H}_t] \\ \text{s.t.} \quad & \mathbb{P}(S_{\text{seq}} = 1 | \mathcal{H}_t, \mathbf{a}) \geq 1 - \epsilon. \end{aligned} \quad (7)$$

The constraint concerns the joint sequence event; multiplying marginal success probabilities would silently assume conditional independence across a reused receiver. The exposure $E = \int u(t) dt$ (ng-min/ml) is only a proxy: delivered mass requires calibrated flow $Q(t)$ via $\int Q(t) c_{\text{delivered}}(t) dt$, and molecule numbers further require a delivery model. A rollout model may be a mechanistic state estimator, a learned state model, or a learned residual; its distribution and task-event calibration must use separate data containing the relevant actions and outcomes.

The controller executes the first action of the solution, logs actuation and delivery, and replans after feedback; without a feasible sequence it keeps a reference wait rule. Its waiting-only special case probes once the estimated success probability reaches a level q (Fig. 2, Sec. VI-E). On measured receivers it requires calibrated delivery, prespecified tasks, an action-supported rollout model and joint-event calibration.

V. EXPERIMENTAL DATA AND EVALUATION PROTOCOL

A. Source, Audit, and Pilot Split

All numerical files come from the paper-linked `MiJan/LFNS_MSB` repository at pinned commit `5c917ab`

TABLE I

SINGLE-CELL TRAJECTORIES PER PROTOCOL AND FGF2 COMMAND AMPLITUDE (NG/ML); FRAMES PER CELL AT 2-MIN SPACING. DAGGERED MIXED BLOCKS ARE THE INSPECTED PILOT TEST; THE PILOT USED ONLY THE 2.5 AND 250 NG/ML BLOCKS.

Protocol	Role	0.25	2.5	25	250	Frames
Sustained	Train	64	59	54	71	79
3/20 pulses	Train	—	70	58	79	79
Single 5 min	Val.	105	88	92	85	69
Single 10 min	Test A	56	67	64	61	69
Single 60 min	Test B	—	66	63	38	69
Mixed pulses	Test C	87	78 [†]	84	76 [†]	130

[13]; the associated data record is cited for provenance [14]. Every file used here matches its Git blob identifier. Commands follow the authors’ model XML and pulse annotations with half-open intervals: 3/20 pulses are 3 min on every 23 min, single pulses start at 0, and mixed pulses span $[1, 4)$, $[24, 54)$ and $[114, 119)$ min. For the mixed protocol we reconstructed the 10–30 min per-cell median normalization from unnormalized exports (maximum discrepancy 1.4×10^{-4} , export rounding); other conditions retain author preprocessing. The pilot trained on the sustained and 3/20 blocks at 2.5 and 250 ng/ml, validated on single 5-min pulses, and tested on mixed pulses (606 cells, 53,998 observations). That test has since been inspected and serves only as context.

B. Pre-Registered Expansion

Before downloading any newly used response file, we froze an analysis protocol (roles, schedules, features, penalty grid $\{10^{-8}, \dots, 10^4\}$, horizons, metrics, hypotheses and data rules) and recorded its SHA-256 hash; the analysis refuses to run if the protocol changes, and it was run once. The 56 additional files all matched their upstream blob identifiers. Table I lists the 22 conditions. Training uses the sustained and 3/20 blocks at all concentrations, validation the single 5-min pulses. Test A is a new protocol (single 10-min pulse, timing from the XML). Test B is the single 60-min pulse; no XML input exists, so its $[0, 60)$ timing was inferred from the authors’ B3 simulations, which match their sustained simulations through 60 min and diverge by 64–78 min, with a pre-registered ± 1 min onset sensitivity; their simulation summary later confirmed this timing. Test C uses the mixed protocol at new concentrations; raw file names show it was recorded as further channels of the inspected experiment, so it shares that session. The primary hypotheses compare causal history (H1) and history plus clock (H2) with the IR-plus-clock comparator at 10 min on Test A. All cells passed the rules (uniform 2-min grids, finite values). For 0.25 ng/ml the exported “unnormalized” file is identical to the normalized one, so its normalization check is uninformative; at 25 ng/ml the discrepancy is 1.4×10^{-4} .

C. B3 Execution and Reference Protocol

B3 is executed by parsing the authors’ model, initial-value and measurement files, with the posterior order taken from

TABLE II

PILOT CONDITION-EQUAL RMSE (NORMALIZED REPORTER UNITS). TEST AT 2 AND 10 MIN, 10-MIN VALIDATION, AND THE 10-MIN TEST SPLIT BY FORECAST ORIGIN AT THE LATEST TRAINING ORIGIN. BOLD: SMALLEST POINT ESTIMATE.

Predictor	2 min	10 min	Val.	≤ 147	> 147
Persistence	0.01588	0.03105	0.02492	0.03698	0.02047
Input only	0.04602	0.04598	0.04400	0.04725	0.04422
Input+response (IR)	0.01554	0.02960	0.02611	0.03364	0.02302
IR + clock	0.01560	0.02923	0.02648	0.03354	0.02209
IR + slope	0.01586	0.02958	0.02591	0.03386	0.02253
Causal history	0.01539	0.02682	0.02453	0.03060	0.02062

their simulation summary; no equation is retyped and no parameter is refit. An acceptance gate fixed in advance required our posterior-predictive mean to match their exported mixed-protocol predictions within 10^{-3} ; the maximum differences were 7.9×10^{-6} and 8.5×10^{-6} , sample-wise errors equal their added measurement noise (10^{-4}), and conservation laws hold to about 10^{-14} . B3 was fitted to sustained and 3/20 means at 2.5 and 250 ng/ml, and mixed and single 5-min conditions at those doses entered its architecture selection. The M0–M3 comparison was frozen in its own protocol before any B3 forecast was scored; because all roles had been inspected for the learned models, it is a post hoc reference analysis.

D. Scores and Comparators

The score is the square root of the equally weighted condition MSE,

$$\text{RMSE}_{\text{cond}}(h) = \left[\frac{1}{|D|} \sum_{d \in D} \frac{1}{N_{d,h}} \sum_{i \in d} (\hat{y}_{i+h} - y_{i+h})^2 \right]^{1/2}, \quad (8)$$

reported with per-condition directions. Overlapping rows and cells are not independent interventions, and experimental-day identifiers are unavailable, so no significance test or confidence interval is reported. Persistence ($\hat{y}_{t+h} = y_t$) is included because it needs no training.

VI. RESULTS

A. Pilot on the Inspected Test

Table II reproduces the pilot. At 10 min, causal history (0.02682) improves on IR plus clock (0.02923) by 8.25% and on IR by 9.39%; at 2 min the gain over IR is 0.94%. Three caveats qualify this. First, 42.5% of the 10-min test forecasts start after the latest training origin (147 min). Within that range the history gain over IR plus clock is 8.8%, but beyond it persistence (0.02047) is the best predictor, and the 2-min gain is +2.0% within and −1.8% beyond. Second, on the validation protocol history improves on persistence by only 1.6%, and IR plus clock is worse than persistence. Third, all response models select the smallest penalty (10^{-6}), so they are effectively unregularized.

B. Pre-Registered Fresh Tests

Table III reports the frozen analysis. Both primary hypotheses hold: on Test A, causal history reduces the 10-min RMSE

TABLE III

10-MIN CONDITION-EQUAL RMSE. VAL.: SINGLE 5-MIN PULSE; A: SINGLE 10-MIN PULSE (NEW PROTOCOL); B: SINGLE 60-MIN PULSE (NEW PROTOCOL); C: MIXED PULSES AT 0.25 AND 25 NG/ML (NEW CONCENTRATIONS, INSPECTED SESSION). ROWS ABOVE M2 ARE THE PRE-REGISTERED ANALYSIS; M2 AND M3 ARE THE POST HOC B3 REFERENCE. BOTTOM ROWS COUNT CONDITIONS WITH THE LOWER MSE.

10-min predictor	Val.	A	B	C
Persistence	0.02353	0.02287	0.02651	0.02876
Input+response (IR)	0.02535	0.02477	0.02711	0.02839
IR + clock	0.02561	0.02490	0.02709	0.02792
IR + slope	0.02503	0.02442	0.02715	0.02825
Causal history	0.02346	0.02270	0.02568	0.02778
History + clock	0.02367	0.02286	0.02624	0.03021
B3 + feedback (M2)	0.02266	0.02286	0.02447	0.02792
History + B3 (M3)	0.02195	0.02098	0.02439	0.02656
History beats clock	4/4	3/4	3/3	2/2
History beats persistence	1/4	1/4	2/3	1/2
M3 beats M2	3/4	4/4	2/3	2/2

from 0.02490 to 0.02270 (8.8%) and history plus clock to 0.02286 (8.2%), with history ahead of IR plus clock at three of four concentrations. The comparison with persistence is less favorable. The current-state models, trained on two protocols, are all worse than persistence on Tests A and B, and causal history improves on persistence by only 0.7% on Test A (one of four concentrations) and 3.1% on Test B (two of three). On Test B the ordering is unchanged under the ± 1 min onset shifts (history 0.02530 and 0.02614 against persistence 0.02651). On Test C, history and IR plus clock are nearly equal (0.02778 versus 0.02792), and history plus clock degrades to 0.03021: before the latest training origin the current-state models are slightly better (IR 0.03216 versus history 0.03284), whereas beyond it causal history is best and the extrapolated clock terms hurt. At 2 min on Test A, causal history improves on IR plus clock by 1.9% (four of four concentrations). Penalty selection again reaches the grid edge (10^{-8} for every 10-min response model).

C. Leave-One-Protocol-Out Robustness

The pre-registered robustness analysis retrains on five protocol groups and tests on the sixth, with penalties fixed at the primary selection (Table IV). History features, with or without clock terms, give the lowest RMSE in all six folds, improving on persistence by 5.8–13.6%. In particular, the 3/20 fold improves from 0.04076 (persistence) and 0.04306 (IR plus clock) to 0.03521. Compared with Table III, this indicates that the history advantage over persistence depends on how diverse the training stimulation is; it remains a protocol-level, single-session-per-protocol result.

D. Mechanistic Reference and Nested History Residual

Table III also lists the B3 forecasters ($\sigma = 0.005$ at both horizons). Without feedback, the B3 posterior-mean forecast is far worse (0.03819 on Test A). With matched feedback, M2 beats IR plus clock (0.02286 versus 0.02490) but only equals persistence (0.02287) and is marginally behind causal history (0.02270). Adding the history features to the M2

TABLE IV
LEAVE-ONE-PROTOCOL-OUT 10-MIN RMSE; INSPECTED MIXED BLOCKS
EXCLUDED FROM TRAINING.

Held-out protocol	Persist.	+ clock	History	Hist.+clock
Sustained	0.02183	0.02135	0.02001	0.02000
3/20 pulses	0.04076	0.04306	0.03521	0.03583
Single 5 min	0.02353	0.02315	0.02182	0.02187
Single 10 min	0.02287	0.02247	0.02130	0.02133
Single 60 min	0.02651	0.02750	0.02497	0.02498
Mixed, new conc.	0.02876	0.02743	0.02699	0.02734

increment (M3) gives the lowest RMSE in every role at both horizons: 0.02098 on Test A, 8.2% below M2 and 8.3% below persistence, ahead of M2 at four of four concentrations; on Tests B and C the reductions relative to M2 are 0.3% and 4.9%. Alone, neither the mechanistic estimator nor the history summary improves on persistence by more than 0.7% on the unseen protocol; only their combination does. This is a reference comparison on inspected roles, not a fresh test.

E. In-Silico Readiness and Waiting on B3

Readiness and waiting policies were then simulated on B3 under a protocol frozen before running (Fig. 3). A 3- or 30-min command at 25 ng/ml ends at time 0 and a 5-min probe follows after a wait; the task is an incremental response of at least half the same sample’s response on an unstimulated receiver within 20 min. Each posterior sample acts in turn as the plant, observed every 2 min with noise (s.d. 0.005), while controllers use the other 999 samples. A first version requiring 80% of the naive response was infeasible for every sample within 120 min (largest recovered fraction 0.79) and is reported unchanged in the package; the version shown changes only that threshold and two rules based on baseline return, which never occurred (a reset policy and the history comparison). Readiness rose only after about 60 min and reached 0.5 at 78 and 88 min after short and long histories, so the candidate waits of 6–48 min never succeeded. Although the reporter never returned within 0.01 of baseline, readiness depended almost only on its current value: nearly all states below a rise of 0.045 were ready and almost none above 0.050, for both histories (mean difference 0.6 percentage points). Weighting the other samples by all observations and probing at $q = 0.9$ (Fig. 2) achieved 90.0% success with a mean failure-penalized completion time of 113.8 min, against 121.8 min for the shortest fixed wait with at least that success (100 min, 91.3%). Using only the current observation gave 75.6% at 119.7 min, a half-decay reset 2.2%, and an oracle 99.3% at 104.9 min. Because readiness is nearly Markov in the reporter, the history-aware advantage here comes from filtering noisy observations rather than hidden memory; the controllers share the plant’s model structure, and none of this is evidence about measured cells.

F. Command-Only Models and Mechanistic Reference

For population means without reporter feedback, a positive single-lag model (validation time constant 13.16 min) and a ridge filter bank with scales 1–128 min were fit on the pilot

TABLE V
POPULATION-MEAN OPEN-LOOP RMSE ON 0–180 MIN (PILOT SPLIT).

Population-mean model	2.5 ng/ml	250 ng/ml
Positive single lag	0.05395	0.05675
Stable filter bank	0.02071	0.02245
Training-mean constant	0.02091	0.02306
Author B3 reference	0.01146	0.01362

split. On 0–180 min (Table V), the heavily regularized bank improves on a constant training mean by only 0.9% and 2.7%, and the single lag is worse than the constant; the mean of the original authors’ 1,000 B3 predictive samples is much closer. B3 used mixed and single-pulse conditions in its architecture selection, so it is a selection-exposed reference, not a blind benchmark, and population-mean errors are not comparable with the individual forecasts above. After the long pulse the mean reporter still rises from 54 to 66 min with zero command, which a nonnegative single lag cannot reproduce; the source biology reports such washout behavior [1].

G. Exploratory Matched History

Pairing early washout after the short mixed pulse (6–12 min) with later washout after the long pulse (56–102 min) at similar reporter values (tolerance 0.005), the later-minus-earlier 10-min target is positive at 2.5 ng/ml (60/78 cells, +0.0144; slope-matched 14/78, +0.0202) but negative at 250 ng/ml, shrinking to −0.0012 when slope-matched (17/76). These post hoc, post-stimulation matches identify no causal history effect.

VII. DISCUSSION AND REQUIRED VALIDATION

The question is molecular because the task involves non-negative chemical actuation, delivery and washout dynamics, a nonlinear receiver, and history-dependent availability: waiting is a control action with a time cost, and exposure has its own cost. Fresh protocols confirm that history features beat a current-state-plus-clock model; against persistence they help mainly when combined with a mechanistic estimator or when training covers diverse stimulation, and the long mixed protocol exposes clock extrapolation. In B3, readiness after 25 ng/ml commands takes over an hour and is nearly determined by the current reporter, so the registered candidate waits should be extended and the expected value of history lies in state estimation under noise; whether measured cells carry history beyond the reporter remains open. None of this separates delivery memory from intracellular memory or establishes task reliability.

A draft validation registration accompanies the package: 3- versus 30-min preconditioning randomized per chamber/session, a common 5-min probe at a fixed protocol clock (90 or 120 min) after 6–48 min waits, and matched controls, with uncalibrated candidate doses and times. The primary comparison is a mechanistic state estimator plus a history residual against the same estimator with identical feedback (group-level Brier score of the next task event); the primary endpoint is the failure-penalized completion time

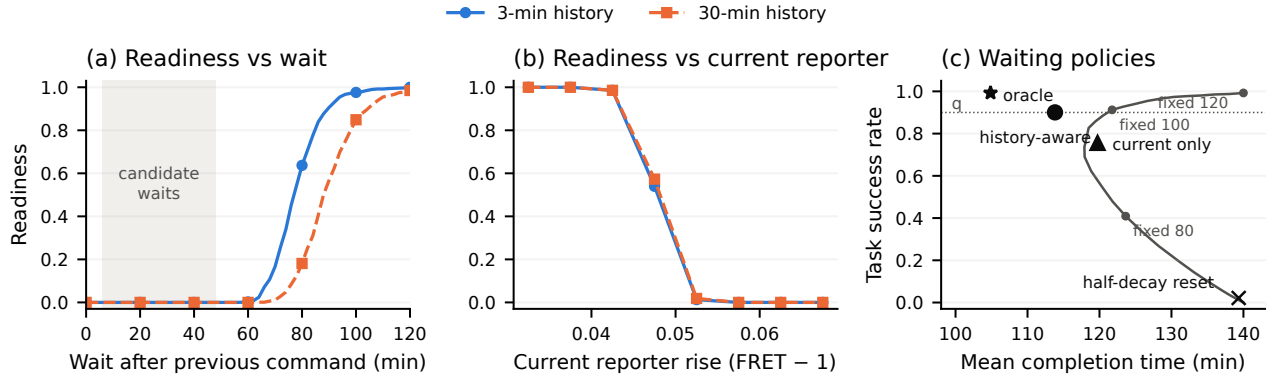


Fig. 3. In-silico readiness on the authors' B3 model (25 ng/ml commands; task: probe response $\geq 50\%$ of the naive response). (a) Fraction of posterior samples ready versus wait after a 3- or 30-min command. (b) Readiness versus the current noise-free reporter rise. (c) Success rate versus mean failure-penalized completion time for waiting policies; the gray curve traces fixed waits, the dotted line the 0.9 target. Simulation of one model structure, not measured cells.

$ST + (1 - S)D$ of two ordered tasks completed by the same eligible cells, starting with adaptive waiting. Sample sizes count independent days or chambers, never cells: zero-failure support for 0.95 sequence reliability needs 59 independent sequences. A targeted audit of 18 neighboring works, with recorded full-text gaps, supports no priority claim. If history adds no independent-day benefit, or a mechanistic estimator reaches the same reliability and cost, the claim should be narrowed.

VIII. CONCLUSION

On experimental FGF2–ERK trajectories, causal-history features improved 10-min prediction over a current-state-plus-clock model in a pilot and in a pre-registered test on unseen protocols, and combined with a matched-feedback execution of the authors' B3 model they beat naive persistence by 8.3% in a post hoc comparison. In simulation on B3, task readiness tracked the current reporter, and a history-aware waiting policy met a 90% reliability target faster than fixed waiting by filtering noisy observations. Demonstrating such benefits for measured receivers requires calibrated delivery, prespecified tasks, supported interventions and new independent-session experiments.

DATA AND CODE AVAILABILITY

The package contains frozen results, protocols with hash records, the B3 execution, source manifests and verification scripts; one command re-runs the analysis and checks the reported values against a frozen reference. Sources are credited in [1], [13], [14]; author files keep their attribution and receive no new license.

REFERENCES

- [1] Y. Blum, J. Mikelson, M. Dobrzyński, H. Ryu, M.-A. Jacques, N. L. Jeon, M. Khamash, and O. Pertz, "Temporal perturbation of ERK dynamics reveals network architecture of FGF2/MAPK signaling," *Molecular Systems Biology*, vol. 15, no. 11, p. e8947, 2019, doi: 10.15252/msb.20198947.
- [2] H. Ryu, M. Chung, M. Dobrzyński, D. Fey, Y. Blum, S. S. Lee, M. Peter, B. N. Kholodenko, N. L. Jeon, and O. Pertz, "Frequency modulation of ERK activation dynamics rewires cell fate," *Molecular Systems Biology*, vol. 11, p. 838, 2015, doi: 10.15252/msb.20156458.
- [3] L. Postiglione, S. Napolitano, E. Pedone, D. L. Rocca, F. Aulicino, M. Santorelli, B. Tumaini, L. Marucci, and D. di Bernardo, "Regulation of gene expression and signaling pathway activity in mammalian cells by automated microfluidics feedback control," *ACS Synthetic Biology*, vol. 7, no. 11, pp. 2558–2565, 2018, doi: 10.1021/acssynbio.8b00235.
- [4] J.-B. Lugagne, C. M. Blassick, and M. J. Dunlop, "Deep model predictive control of gene expression in thousands of single cells," *Nature Communications*, vol. 15, p. 2148, 2024, doi: 10.1038/s41467-024-46361-1.
- [5] Y. Cheng, W. Chen, and B. Ai, "Building semantic communication system via molecules: An end-to-end training approach," *arXiv preprint arXiv:2404.09595v1*, 2024, <https://arxiv.org/abs/2404.09595v1>.
- [6] H. Cai and O. B. Akan, "Semantic learning for molecular communication in internet of bio-nano things," *arXiv preprint arXiv:2502.08426v2*, 2025, <https://arxiv.org/abs/2502.08426v2>.
- [7] L. Dong, "Interpretable reaction–diffusion learning for molecular communication via graph Kolmogorov–Arnold networks," *IEEE Transactions on Molecular, Biological, and Multi-Scale Communications*, vol. 12, pp. 1056–1076, 2026, doi: 10.1109/TMBMC.2026.3731398.
- [8] N. Ntetsikas, S. Kyriakoudi, A. Kirmizis, I. Krikidis, I. F. Akyildiz, and M. Lestas, "Noise characterization and robust signal detection in yeast pheromone molecular communication," *IEEE Transactions on Molecular, Biological, and Multi-Scale Communications*, vol. 11, no. 2, pp. 218–227, 2025, doi: 10.1109/TMBMC.2025.3554640.
- [9] M. G. Durmaz, N. Tulluk, R. D. Aksoy, H. B. Yilmaz, B. Yang, A. Wipat, A. E. Pusane, G. Misirlı, and T. Tugcu, "BioRxToolbox: a computational framework to streamline genetic circuit design in molecular data communications," *Synthetic Biology*, vol. 9, no. 1, p. ysae015, 2024, doi: 10.1093/synbio/ysae015.
- [10] M. T. Barros and S. Dey, "Feed-forward and feedback control in astrocytes for Ca^{2+} -based molecular communications nanonetworks," *IEEE/ACM Transactions on Computational Biology and Bioinformatics*, vol. 17, no. 4, pp. 1174–1186, 2020, doi: 10.1109/TCBB.2018.2887222.
- [11] L. F. Borges, M. T. Barros, and M. Nogueira, "Cell signaling error control for reliable molecular communications," *Frontiers in Communications and Networks*, vol. 5, p. 1332379, 2024, doi: 10.3389/fr-cmn.2024.1332379.
- [12] M. Scherer, L. Brand, L. Wolf, T. tom Dieck, M. Schäfer, S. Lotter, A. Burkovski, H. Sticht, R. Schober, and K. Castiglione, "Closed-loop long-term experimental molecular communication system," *IEEE Transactions on Molecular, Biological, and Multi-Scale Communications*, vol. 12, pp. 22–41, 2026, author version: arXiv:2502.00831v2, doi: 10.1109/TMBMC.2025.3615924.
- [13] LFNS_MSB contributors, "LFNS_MSB: Paper-linked experimental time courses and inference exports," GitHub, branch MSB_version, 2019, pinned commit 5c917ab. https://github.com/Mijan/LFNS_MSB.
- [14] M. Dobrzyński, "Temporal perturbation of Erk dynamics reveals network architecture of FGF2-MAPK signaling," Mendeley Data, version 2, 2020, contributor: Y. Blum. doi: 10.17632/ccnxn84w8z.2.